

FRANK EBELEDIKE

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SUMMARY

Software engineer with 4+ years of development experience, now focused on DevOps and cloud infrastructure. With hands-on experience in CI/CD, Terraform, Docker, Kubernetes and AWS, I am driven by the belief that great applications are built on optimized infrastructure, I build production apps as well as engineer the systems that power them.

EDUCATION

Masters of Science in Information Technology

August 2025 - June 2027

Carnegie Mellon University, Africa

- **Courses:** Cloud Computing, DevOps, AIOps, Networking Lab, System Software Eng., Data Structures & Algorithms, Software Architecture

Bachelor of Engineering in Computer Engineering

March 2019 - September 2024

Federal University of Technology, Minna

- Graduated with First Class, Among top 1% of the Class of 2024
- **Thesis:** Intelligent Sensor-Based Cyber-Physical Intrusion Detection System to protect Telecommunication Infrastructure

TECHNICAL SKILLS

Languages. Python, C/C++, Java, Kotlin, Swift, NodeJs, SQL, Bash/Shell scripting, HTML, XML, CSS.

DevOps/Cloud Docker, Kubernetes, SageMaker, Jenkins, Ansible, Terraform, Vagrant, GitHub Actions, Azure Pipelines, Databrick, Prometheus, Grafana, AWS.

Tools/Databases Flutter, Firebase, MongoDB, MySQL, Postman, Linux, ApacheSpark, Apache HBase.

Version Control & Registries Git, GitHub, Google Artifact Registry (GAR), Azure Container Registry (ACR).

PROFESSIONAL EXPERIENCE

AWS Cloud Engineer

November 2025 - Present

AfryScripta - WiNEST Research, Kigali, Rwanda

- Served as the infrastructure administrator, optimizing AWS account security configurations, IAM roles, and resource utilization.
- Designed and deployed a serverless architecture using AWS AppSync, DynamoDB, S3, and Cognito, serving 500+ users with 99.9% availability and zero infrastructure management overhead.
- Automated infrastructure provisioning using AWS CDK and Amplify CI/CD pipelines with sandbox environments, enabling reproducible deployments across dev and production with zero-downtime releases.
- Engineered a secure IAM and Cognito-based RBAC system with pre-signed S3 uploads, group-based API authorization, and DynamoDB TTL-based session management, reducing unauthorized access surface by 100%.

Software Engineer

November 2024 - Present

TMR International Hospital, Uganda (Remote, Part-time)

- Designed and maintained Azure Pipelines CI/CD workflows to automate build, test, and distribution of 3 production healthcare apps (Patient, Doctor, Rider) to testers via automated delivery, reducing manual release overhead.
- Instrumented apps with Firebase Performance Monitoring and Crashlytics to track real-time performance metrics and crash reports, enabling data-driven fixes that contributed to a 12% reduction in app load times.
- Delivered cross-platform Flutter apps on Android and iOS serving 1,200+ active users, achieving a 90% user satisfaction rating based on app store reviews.

PROJECTS

Cloud Native Web Service Developed a distributed ETL pipeline using Spark and Databricks on GCP Dataproc which is optimized based on cost and performance. Exported processed data into AWS EBS volume and mounted data as a PersistentVolume to pod in Kubernetes cluster

Intelligent Fare Prediction & Voice Assistant Architected and deployed a ride fare prediction platform on GCP using XGBoost on Cloud Run, served via Vertex AI and App Engine. Integrated a voice-enabled AI assistant backed by a RAG pipeline and LangGraph agentic workflow for real-time ride queries and location-aware recommendations.

Cloud Native Chat App (Multi-Cloud Deployment) Containerized services with Docker and managed image versioning across GAR and ACR. Deployed to Kubernetes with Helm charts; configured Horizontal Pod Autoscaler for automatic scaling under load and NginX Ingress for traffic routing on GCP. Implemented Azure Front Door for global routing and failover, improving application availability and reducing latency for geographically distributed users.

MSB Elastic Cloud Infrastructure Provisioned a fully elastic AWS infrastructure using Terraform, including ELB and Auto Scaling Groups, enabling dynamic resource allocation based on workload demand. Analyzed traffic and workload patterns to define elasticity policies that balanced performance SLAs against budget constraints, reducing over-provisioning costs

PetClinic Infrastructure Deployed the PetClinic application on a 3-node Kubernetes cluster with full lifecycle management including rolling updates and health checks. Integrated Nagios Core for infrastructure monitoring across nodes and implemented Ansible Vault for secrets management, improving deployment security posture.

Flutter Intl Translate - OpenSource Built and open-sourced a JSON/ARB translation automation tool for Flutter internationalization workflows, adopted by 170+ developers on the VS Code Marketplace.

AWARDS & CERTIFICATIONS

- **Kubernetes and Cloud Native Essentials**
- **AWS Cloud Practitioner Essentials**
- **ITU WTSA-24 Hackathon (2024):** Built WiNEST Chatbot during 40-hour coding challenge in New Delhi, India.
- **ITU Build-a-Thon Challenge (2022):** 2nd Runner-up and Grand Challenge Finalist in ITU AI/ML in 5G Challenge.
- **Certificate of Excellence for Outstanding Academic Performance** (2018/2019), (2019/2020)